A Multispectral Texture-Feature Machine Learning Framework for Cotton Bacterial Blight Severity Classification

Authors: Hafiz Muhammad Ijaz (ORCID: 0009-0002-5249-0622), Muhammad Asadullah (corresponding author)
Affiliation: Department of Information Technology, The Islamia University of Bahawalpur, Pakistan

1. Description

This project classifies the severity of cotton bacterial blight (*Xanthomonas campestris* pv. *malvacearum*) from leaf images with a feature-based machine-vision pipeline:

1. Each cotton leaf image is preprocessed: resizing, grayscale conversion (CVIPtools), and contrast enhancement with noise reduction.
2. Regions of interest (ROIs) are segmented at different pixel sizes (16 × 16 up to 256 × 256).
3. **55 candidate features** (first-order histogram, GLCM/Haralick texture, binary shape, and rotation-scale-translation (RST) invariant features) are extracted from each ROI with CVIPtools.
4. **Correlation-based Feature Selection (CFS)** reduces the 55 features to a compact subset (19 features plus the class label = a 20-attribute file).
5. **Twelve machine learning classifiers** and **four deep learning models** are trained and evaluated in Weka with **10-fold cross-validation**.

The task is a three-class severity problem:

Class	Meaning
Low infection	Small affected leaf area, early and mild symptoms
Normal infection	Moderate, clearly visible portion of the leaf shows blight symptoms
High infection	Extensive necrotic and water-soaked lesions

2. Dataset Information

Item	Details
Name	A Multispectral Texture-Feature Machine Learning Framework for Cotton Bacterial Blight Severity Classification
Repository	GitHub: https://github.com/ijazhafiz7-gif/Cotton-Bacterial-Blight-Classification
DOI	None assigned (see the repository link above)
Origin	Cotton leaf images captured under natural field lighting and checked manually before processing [add location, season and camera details]
Ethics approval	Not applicable: plant images, no human or animal subjects
Modality	RGB photographs of cotton leaves, provided as 256 × 256 pixel JPEG images
Size	900 leaf images (300 per class)
Annotation	Each image was manually assigned to Low, Normal or High infection before image processing
License	Creative Commons Attribution 4.0 International (CC BY 4.0)

Images per category:

Category	Images	Proportion
Low infection	300	33.33%
Normal infection	300	33.33%
High infection	300	33.33%
Total	**900**	**100%**

The dataset is balanced on purpose, so that accuracy and F-measure are not dominated by an over-represented class. Feature extraction produces 900 ROI instances per class (2,700 instances in total), which are the instances used in cross-validation.

Data layout

```
cotton_blight_dataset/
├── README.md
├── manifest.csv
├── Low_Infection/       (300 images: 1.jpg ... 300.jpg)
├── Normal_Infection/    (300 images: 1.jpg ... 300.jpg)
├── High_Infection/      (300 images: 1.jpg ... 300.jpg)
├── features/
│   └── Cotton_Balanced_900.arff
└── results/
    └── 12 Weka output files (.docx), one per classifier
```

All images are 256 × 256 pixel RGB JPEG files. File names repeat across the three class folders (each has 1.jpg to 300.jpg), so use the `image_id` column of `manifest.csv` as the unique identifier.

`manifest.csv` has one row per image (900 rows):

Column	Meaning
`path`	Relative path of the image inside the dataset folder
`image_id`	Unique image ID (class name + number, e.g. `Low_1`)
`class`	Infection severity: `Low`, `Normal` or `High`

The ARFF file `features/Cotton_Balanced_900.arff` has 2,700 rows (900 per class). Each row holds the **19 selected features** and the class label (20 attributes). The class values in the file are `Lowinfection`, `Normalinfection` and `HighInfection`:

Feature group	Selected features	Count
Shape / binary	Object row coordinate, object column coordinate, area, centroid (row), centroid (column), second moment, perimeter, Euler number	8
Projection	Projections 1-11 (normalized row and column projection profiles)	11
Total		**19**

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3. Code Information

File / tool	Purpose	Main outputs
CVIPtools	**Step 1.** Image preprocessing, ROI extraction and extraction of the 55 candidate features	Feature file (ARFF)
Weka (ARFF files)	**Step 2.** CFS feature selection, training and 10-fold cross-validation of 16 classifiers	Accuracy, confusion matrices, per-class metrics
`features/Cotton_Balanced_900.arff`	Selected 20-attribute feature set (2,700 instances) used to train and evaluate all classifiers	
`manifest.csv`	Lists every image with its unique ID and infection class	
`results/*.docx`	Weka output for each of the 12 machine learning classifiers (10-fold cross-validation, 2,700 instances)	Accuracy, per-class metrics, confusion matrix
`README.md`	This file	

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4. Requirements

Software

Software	Version
CVIPtools	3.9
Weka	3.6.2

Computing infrastructure

Component	Specification
Device	Acer TravelMate 8571 laptop
Operating system	Windows 10 Pro, 64-bit (version 20H2, OS build 19042.1706)
CPU	Intel(R) Core(TM)2 Duo CPU U9400 @ 1.40 GHz (x64-based processor)
RAM	6.00 GB
GPU	Not required (CVIPtools and Weka were run on the CPU)

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5. Usage Instructions

1. **Download the dataset** from GitHub (https://github.com/ijazhafiz7-gif/Cotton-Bacterial-Blight-Classification) and unzip it.
2. **Open CVIPtools** and load the images. Resize, convert to grayscale, and apply contrast enhancement and noise reduction.
3. **Segment ROIs** at the sizes used in the study (16 × 16 and 256 × 256 pixels).
4. **Extract features** (histogram, texture, binary and RST invariant) for each ROI and save them in ARFF format.
5. **Open the ARFF file in Weka.** Apply Correlation-based Feature Selection (CFS) to keep the 20-attribute subset.
6. **Run the classifiers** in the Weka Classify tab with *10-fold cross-validation* and note the weighted-average results.

Fixed settings

Setting	Value
Images	900 (300 per class)
Resized image / ROI sizes	256 × 256 and 16 × 16 pixels
Candidate features	55
Selected features	19 + class label = 20 attributes
Feature selection	Correlation-based Feature Selection (CFS)
Evaluation	10-fold cross-validation
Machine learning classifiers	Lazy.IB1, Lazy.IBK, Trees.RandomTree, Meta.RandomCommittee, Trees.RandomForest, Rules.NNge, Meta.RotationForest, Meta.Decorate, Trees.SimpleCart, Trees.J48, Trees.J48graft, Trees.BFTree
Deep learning models	CNN, LSTM, Bi-LSTM, MLP (on feature vectors, not raw pixels)
Classifier parameters	Weka 3.6.2 settings as recorded in the result files in `results/`, e.g. IBk -K 1; RandomForest -I 10 -K 0 -S 1; RandomTree -K 0 -M 1.0 -S 1; J48 -C 0.25 -M 2; RandomCommittee -S 1 -I 10
Random seed	1 (as recorded in the Weka settings, `-S 1`)

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6. Methodology

1. **Pre-processing:**
    1. Resizing: every image is resized to a fixed size so that ROI segments are comparable.
    2. Grayscale conversion: a luminance-weighted transformation of the R, G and B channels (CVIPtools).

    3. Enhancement: contrast enhancement and noise-reduction filters, to suppress background noise and normalize brightness.
2. **ROI segmentation:** the infected leaf region is separated from the background. ROIs are produced at several pixel resolutions, from 16 × 16 to 256 × 256, to test how ROI size affects accuracy. Small ROIs emphasize local texture and early lesions, large ROIs keep more spatial context.
3. **Feature extraction (CVIPtools):** 55 features per ROI from four families:
    - First-order histogram features (mean, standard deviation, skewness, energy, entropy).
    - Second-order GLCM (Haralick) texture features.
    - Binary shape features (area, centroid, moments, perimeter, Euler number, thinness ratio, aspect ratio, projections).
    - Seven RST invariant moments.
4. **Feature selection:** CFS keeps features that correlate strongly with the class and weakly with each other. The merit of a subset S with k features is Merit_S = k · r_cf / sqrt(k + k(k-1) · r_ff), where r_cf is the mean feature-class correlation and r_ff the mean feature-feature correlation. The final set is dominated by shape descriptors and projection profiles.
5. **Classification:** 12 machine learning classifiers and 4 deep learning models run in Weka on the 20-attribute feature file.
6. **Evaluation:** 10-fold cross-validation. The data is split into ten nearly equal subsets, nine are used for training and one for testing, ten times, and the scores are averaged.

Evaluation method

- **Per-class and weighted-average results** for Low, Normal and High infection.
- **Confusion matrices** for the best classifiers.
- **Classifier comparison:** all 12 machine learning classifiers on the 16 × 16 ROI feature set, plus deep learning models and a Naive Bayes baseline.
- **ROI size analysis:** 256 × 256 versus 16 × 16 pixels.
- **Comparison with published methods:** Anwar, Rahim Kolachi et al. (2023), Kumar Gudela et al. (2023), Islam et al. (2023), M. Shao et al. (2023), Y. Shao et al. (2024), Akbar et al. (2024), Anwar, Soomro et al. (2023) and Suleman Memon et al. (2022) (Table 10 of the article).

Assessment metrics

Metric	Why it is used
Accuracy	Overall proportion of correctly classified instances.
TP rate (recall / sensitivity)	Share of each class that is correctly recognized.
FP rate	How often samples of other classes are wrongly assigned to a class.
Precision	Of the instances predicted as a class, the fraction that truly belong to it.
F-measure	Harmonic mean of precision and recall.
ROC area	Threshold-independent discrimination ability.
Cohen's kappa	Agreement beyond chance.
MAE / RAE / RMSE	Size of the prediction error.
MCC	Single balanced score for multi-class results (Gorodkin's generalization).
Confusion matrix	Shows which classes are confused with each other.

Main results (Lazy.IB1, ROI 16 × 16, 10-fold cross-validation)

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area
Low infection	0.913	0.037	0.925	0.913	0.919	0.938
Normal infection	0.940	0.038	0.926	0.940	0.933	0.951
High infection	0.947	0.025	0.950	0.947	0.948	0.961
Weighted average	**0.933**	**0.033**	**0.933**	**0.933**	**0.933**	**0.950**

Confusion matrix (rows = actual, columns = predicted):

Actual \ Predicted	Low	Normal	High	Total
Low	822	50	28	900
Normal	37	846	17	900
High	30	18	852	900
Total	889	914	897	2,700

All machine learning classifiers (ROI 16 × 16, weighted averages)

Classifier	Kappa	MAE	RAE (%)	TP Rate	FP Rate	F-Score	MCC	ROC	Accuracy (%)
IB1	0.900	0.0444	10.00	0.933	0.033	0.933	0.900	0.950	**93.33**
IBK	0.898	0.0451	10.16	0.932	0.034	0.932	0.898	0.970	93.22
RandomTree	0.895	0.0468	10.53	0.930	0.035	0.930	0.895	0.947	93.00
Random Committee	0.894	0.0471	10.60	0.930	0.035	0.930	0.894	0.974	92.96
RandomForest	0.893	0.0645	14.52	0.929	0.036	0.928	0.893	0.976	92.85
NNge	0.890	0.0489	11.00	0.927	0.037	0.927	0.890	0.945	92.67
Rotation Forest	0.863	0.0900	20.24	0.909	0.046	0.909	0.863	0.979	90.89
Decorate	0.815	0.1505	33.86	0.877	0.062	0.876	0.815	0.957	87.67
SimpleCart	0.798	0.1043	23.46	0.866	0.067	0.865	0.799	0.947	86.56
J48	0.782	0.1113	25.05	0.855	0.073	0.854	0.783	0.946	85.48
J48graft	0.782	0.1113	25.05	0.855	0.073	0.854	0.783	0.946	85.48
BFTree	0.748	0.1225	27.56	0.832	0.084	0.831	0.749	0.942	83.19

Among the deep learning models, CNN and MLP reached only about 33.33% accuracy, and a Naive Bayes baseline reached 63.75%.

Effect of ROI size

ROI size	Lazy.IB1 (%)	Lazy.IBK (%)	Best ensemble classifier (%)
256 × 256	93.22	93.22	92.96 (Random Committee)
16 × 16	**93.33**	93.22	92.93 (Random Committee)

7. Conclusions

- A compact feature-based pipeline (texture, histogram, binary and RST features, reduced by CFS to 20 attributes) separates Low, Normal and High bacterial blight severity with high accuracy.
- The instance-based **Lazy.IB1** classifier gave the best result: **93.33% weighted-average accuracy** on the 16 × 16 ROI feature set, ahead of Lazy.IBK (93.22%), Random Tree (93.00%), Random Committee (92.96%) and Random Forest (92.85%).
- Every one of the twelve machine learning classifiers outperformed the CNN and MLP models, which reached only about 33.33% on the same pre-extracted features.
- Most errors occur between Low and Normal infection, which have the most similar textures. High infection is recognized best (TP rate 0.947).
- The 16 × 16 ROI performed slightly better than 256 × 256 (93.33% versus 93.22%), so the features carry disease information at both scales.
- The approach is interpretable and light enough for low-cost decision support in agriculture. A formal timing benchmark was not performed.

8. Limitations

- **Three severity levels only:** completely healthy leaves are not included in the classification.
- **Limited data source:** images come from a relatively small number of fields and locations with specific lighting, so results may differ for other climates, cotton varieties, fields or cameras.

- **Visible symptoms only:** other foliar diseases can look similar, and temperature, humidity and rainfall were not considered.
- **No independent test set:** results come from 10-fold cross-validation on one dataset, with no external validation.
- **No timing benchmark:** claims about computational efficiency are based on model and input size, not measured run time.

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9. Citations

Code / paper:

```
Ijaz, H. M., & Asadullah, M. (2026). A Multispectral Texture-Feature Machine Learning
Framework for Cotton Bacterial Blight Severity Classification. [Journal name].
Data and code: https://github.com/ijazhafiz7-gif/Cotton-Bacterial-Blight-Classification
```

Dataset:

```
Ijaz, H. M., & Asadullah, M. (2026). A Multispectral Texture-Feature Machine Learning Framework for Cotton
Bacterial Blight Severity Classification [Dataset]. GitHub. https://github.com/ijazhafiz7-gif/Cotton-
Bacterial-Blight-Classification
```

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10. License

- **README and feature files:** CC BY 4.0, same as the dataset.
- **Dataset:** Creative Commons Attribution 4.0 International (CC BY 4.0).